

CLINICAL RESEARCH

Clinical Trials

Secondary Prevention With Folic Acid: Effects on Clinical Outcomes

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OBJECTIVES	We sought to conduct a randomized trial with folic acid 0.5 mg/day in a patient population with stable coronary artery disease (CAD).
BACKGROUND	Folic acid has favorable effects on vascular endothelium and lowers plasma homocysteine levels. In addition, homocysteine appears to be an independent risk factor for atherosclerotic disease. However, the value of folic acid in secondary prevention had seldom been tested.
METHODS	In this open-label study, 593 patients were included; 300 were randomized to folic acid and 293 served as controls. Mean follow-up time was 24 months. At baseline all patients had been on statin therapy for a mean of 3.2 years.
RESULTS	In patients treated with folic acid, plasma homocysteine levels decreased by 18%, from 12.0 ± 4.8 to 9.4 ± 3.5 $\mu\text{mol/l}$, whereas these levels remained unaffected in the control group ($p < 0.001$ between groups). The primary end point (all-cause mortality and a composite of vascular events) was encountered in 31 (10.3%) patients in the folic acid group and in 28 (9.6%) patients in the control group (relative risk 1.05; 95% confidence interval: 0.63 to 1.75). In a multifactorial survival model with adjustments for clinical factors, the most predictive laboratory parameters were, in order of significance, levels of creatinine clearance, plasma fibrinogen, and homocysteine.
CONCLUSIONS	Within two years, folic acid does not seem to reduce clinical end points in patients with stable coronary artery disease (CAD) while on statin treatment. Homocysteine might therefore merely be a modifiable marker of disease. Thus, low-dose folic acid supplementation should be treated with reservation, until more trial outcomes become available. (J Am Coll Cardiol 2003;41:2105–13) © 2003 by the American College of Cardiology Foundation

In patients with coronary artery disease (CAD), homocysteine has been described as an independent predictor of mortality (1–4). A mild to moderate elevation of homocysteine is common and is strongly related to folate status (5). Nevertheless, the exact mechanism of the deleterious effects of homocysteine on the progression of atherosclerotic disease is still unknown. It has been suggested that it involves endothelial damage, increased adhesiveness of platelets, and disturbance of the clotting cascade (6).

Simple and inexpensive therapy with folic acid reduces plasma homocysteine levels and normalizes endothelial function (7,8). Folic acid dose as low as 0.5 mg/day is sufficient to effectively lower plasma homocysteine level (9). Moreover, it has also been demonstrated that normalization of endothelial function by folic acid appears to be independent of its effect on homocysteine (10). In addition, one clinical trial with folic acid-containing multivitamins has shown to prevent restenosis following percutaneous coronary intervention (PCI) (11). Yet, up till now, the folic acid hypothesis has not been tested in a population with stable

atherosclerotic disease with hard clinical end points such as cardiovascular mortality or morbidity.

METHODS

Patients. Patients with stable CAD were screened for inclusion. Patient history had to include one of the following: myocardial infarction (MI), significant coronary artery lesions (>60%) on coronary angiography, PCI, and/or coronary artery bypass graft surgery (CABG). Patients had to be stable, with no invasive vascular procedures scheduled. Patients were eligible when they were on statin therapy for at least three months. Patients taking any form of vitamin B-containing medication, regularly or sporadic, were considered eligible for this study.

Main exclusion criteria were age below 18 years, history of low vitamin B₁₂ level, treatment for hyperhomocysteinemia, severe renal failure, or any other treatment for renal disease, known hepatic disease, signs and symptoms of severe heart failure (New York Heart Association functional class IV) or any other serious illness that would exclude follow-up time of at least three years.

Study treatment and assessments. For the study, consecutive patients visiting the outpatient department of the cardiology department were screened. As soon as eligibility was confirmed, patients were invited to participate in the study. Written informed consent was obtained. A computer

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Abbreviations and Acronyms

CABG	= coronary artery bypass graft
CAD	= coronary artery disease
CVA	= cerebrovascular accident
HDL-C	= high-density lipoprotein-cholesterol
HS-CRP	= high-selective C-reactive protein
LDL-C	= low-density lipoprotein-cholesterol
MI	= myocardial infarction
PCI	= percutaneous coronary intervention
RR	= relative risk
TC	= total serum cholesterol
TIA	= transient ischemic attack

program randomly allocated patients either to treatment with open-label folic acid 0.5 mg/day or to standard care. Standard care implied the same intensive follow-up and treatment of risk factors. A nurse counselled all patients. In all patients, dosage of statins was increased when necessary. At least one of four goals was meticulously pursued: first, a decrement of low-density lipoprotein cholesterol (LDL-C) of 30% (compared to values before the initiation of statin treatment); second, an LDL-C value of <3 mmol/l; third, an apolipoprotein-B (apoB) value of <1 g/l; and fourth, a decrement of apoB values of 30% compared to prestatin values. During the trial, patients were encouraged to implement and continue dietary measures. Persistent nicotine use was discouraged at regular intervals. Patients were followed for a maximum of 36 months. During the entire study, clinical events were carefully registered. Visits for laboratory examinations were planned at 3, 6, 12, 24, and 36 months, and patients were encouraged to contact by telephone in order to be informed about their cholesterol status. Annual visits with the nurse were scheduled.

The study was conducted in accordance with the Declaration of Helsinki as revised in 1996, and it was performed in a rural area in the vicinity of the city of Goes, called the Bevelanden (province of Zeeland, the Netherlands), from November 1998 to January 2002. The local ethics committee approved the study protocol prior to the start of the trial.

Laboratory procedures. Blood samples were taken in the morning after an overnight fast. All analyses were done in the same hospital laboratory. Total serum cholesterol (TC) and triglyceride concentrations were measured enzymatically (Vitros analyzer, Johnson & Johnson). High-density lipoprotein-cholesterol (HDL-C) fractions were prepared by precipitation from serum of apoB-containing lipoproteins with the use of dextran sulfate and $MgCl_2$ (magnesium chloride). Serum LDL-C was calculated using the Friedewald formula (Friedewald, Clin Chem 1972). Apolipoprotein A-I (apoA-I) and apoB were determined by immunonephelometry on a Beckman array system. Beckman reagents, calibrators, and standards were used. From April 2000, high-selective C-reactive protein (HS-CRP) measurements were done with particle-enhanced turbidimetric immunoassay technique, which provided a sensitivity of 0.5 mg/l and represented the lowest concentration of CRP that

could be distinguished from zero (Dade Behring Dimension Chemical Chemistry System). Plasma fibrinogen levels were determined using the IL-test PT-fibrinogen recombinant method (ACL Futura).

Creatinine clearance was estimated by the Cockcroft-Gault equation (12). Total homocysteine level was measured in plasma with a fluorescence polarization immunoassay on an Imx analyzer. Imx reagents and calibrators were used (Abbott). As synthesis of homocysteine continued in red blood cells after drawing, samples were centrifuged immediately and put on ice. Afterwards, measurements were done within a few hours. Serum folate level was measured with an ion capture assay (Imx) using monoclonal antibodies and was expressed as nmol/l. As recent food intake could influence folate levels, the specimens were taken in a fasting state as well. Special care was taken not to hemolyze the samples, as red-blood-cell-containing samples can give falsely elevated folate levels. Vitamin B₁₂ was measured with an Imx B₁₂ assay based on microparticle enzyme immunoassay technology (Abbott).

Study end points. The primary end point was a composite of vascular events. These events were defined as vascular death (sudden death, fatal recurrent MI, fatal stroke, and other cardiovascular deaths), noncardiovascular death, recurrent MI, or invasive coronary procedures (PCI, CABG), cerebrovascular accident (CVA), or transient ischemic attack (TIA), or any other vascular surgery such as carotid endarterectomy, abdominal aneurysmectomy, or peripheral vascular surgery including limb amputation for vascular reasons. Myocardial infarction was defined according to previously used criteria; two out of three criteria should be positive: chest pain lasting ≥ 30 min, creatine kinase elevation ≥ 2 times the upper limit of normal or new pathological Q waves of ≥ 0.04 -s duration or $\geq 25\%$ of the corresponding R-wave amplitude, both in at least two contiguous leads. The decision to proceed to any coronary vascular procedures such as PCI or CABG was made in other hospitals without knowledge of folic acid substitution. Hospitalization due to increasing anginal complaints with or without troponin elevation, which was determined as "unstable angina pectoris," was recorded but was not considered as a part of the primary end point in this study. Adjudication of all clinical events was performed by an independent end point monitoring committee unaware of treatment arm.

Statistical analysis. All randomized patients fulfilling the eligibility criteria were included in the analysis of time to a major clinical event according to intention-to-treat principle. Continuous data were presented as mean and SD. Continuous data were compared in the two treatment groups using unpaired *t* tests. Categorical data were presented by percentage and count of each category. Treatment comparisons were made using the Fisher exact test or chi-squared test. For correlations, bivariate correlation coefficient was calculated with the Spearman rank. Time-to-event were presented as Kaplan-Meier survival curves and were compared using log-rank tests. For covariate analysis,

Table 1. Baseline Characteristics of 593 Patients Randomly Assigned to Folic Acid (n = 300) or Controls (n = 293)

Patient Characteristic	Folic Acid	Control	p Value
Age (yrs)	64.9 ± 9.9	65.5 ± 9.7	0.42
Male gender (%)	76	80	0.17
BMI (kg/m ²)	26.7 ± 3.4	27.3 ± 3.6	0.026
History of			
MI (%)	59	53	0.08
PCI and/or CABG (%)	58	65	0.06
TIA and/or CVA (%)	6	7	0.45
Peripheral vascular disease (%)	8	9	0.41
Diabetes (%)	7	11	0.09
Use of vitamin B-containing tablets (%)	15	16	0.61
Persistent nicotine use (%)	15	16	0.46
Use of lipid-lowering therapy before randomization (yrs)	3.2 ± 2.4	3.3 ± 2.3	0.55
Mean follow-up time (months)	24.2 ± 10.9	23.2 ± 10.1	0.23
Laboratory Parameters			
At inclusion			
Total cholesterol (mmol/l)	4.7 ± 0.8	4.5 ± 0.7	0.002
LDL-C (mmol/l)	2.7 ± 0.7	2.6 ± 0.6	0.006
HDL-C (mmol/l)	1.12 ± 0.33	1.14 ± 0.30	0.96
Triglycerides (mmol/l)	1.8 ± 0.91	1.8 ± 0.96	0.88
ApoA1 (g/l)	1.35 ± 0.20	1.31 ± 0.20	0.054
ApoB (g/l)	0.93 ± 0.20	0.91 ± 0.18	0.11
Vitamin B ₁₂ (pmol/l)	286 ± 129	294 ± 162	0.51
Homocysteine (μmol/l)	12.0 ± 4.8	12.2 ± 3.8	0.67
Folate (nmol/l)	17 ± 7	15 ± 5	<0.001
Creatinine clearance (ml/min)	78 ± 26	77 ± 24	0.80
Fibrinogen (g/l)	3.98 ± 0.87	4.01 ± 0.82	0.63

BMI = body mass index; CABG = coronary artery bypass graft; CVA = cerebrovascular accident; HDL-C = high-density lipoprotein; LDL-C = low-density lipoprotein; MI = myocardial infarction; PCI = percutaneous coronary intervention; TIA = transient ischemic attack.

the Cox regression analysis was used. The study was powered for a 50% reduction in clinical events on the basis of existing observational data in populations with CAD (1–3). We assumed that the two-year event rate was about 15%. In order to obtain 80% power (5% significance level) to detect a relative risk (RR) of 0.5, we included 300 patients in each treatment group. Analyses were performed using commercially available computer software (SPSS).

RESULTS

Baseline. From November 1998 to September 2001, a total of 593 patients were included in this study. Of these patients, 300 were randomly assigned to folic acid 0.5 mg/day, and 293 patients served as a control group. Twenty eligible patients did not want to participate in the study mainly because of insufficient transport facilities in our rural area. Mean follow-up time was 24 ± 10 months. Baseline characteristics are depicted in Table 1. Manifestations of CAD such as previous MI, PCI, or CABG were evenly distributed between both groups as were other concomitant vascular diseases, such as CVA or TIA or peripheral vascular disease.

Lipid parameters. Mean TC value before treatment was 6.8 mmol/l; mean LDL-C, 4.7 mmol/l; mean HDL-C, 1.1 mmol/l; mean apoA-1, 1.3 g/l; and mean apoB, 1.4 g/l. Laboratory results at baseline are shown in Table 1. As most

patients were already on statin therapy for more than three years, the TC level at inclusion was already well below 5 mmol/l in the majority of patients. During follow-up the mean TC and LDL-C values did not change (data not shown). In both groups, statin treatment was intensified when necessary.

Homocysteine, folic acid, vitamin B₁₂, and creatinine clearance. At baseline, in both groups 15% to 16% of patients used vitamin B-containing medication, sometimes regularly and sometimes only in winter months (Table 1). Most vitamin B users mentioned use of a complex form of vitamin B, which contained both folic acid and pyridoxine. During the study, patients were neither encouraged nor discouraged to continue these vitamin supplements. As stated before, for reason of extrinsic validity, patients using vitamin supplementation on their own initiative were considered eligible for this study. Patients using vitamin B-containing supplements had a higher level of serum folate (19.4 ± 8.2 vs. 15.4 ± 5.5 nmol/l, p < 0.001) and lower level of plasma homocysteine (11.0 ± 3.8 vs. 12.3 ± 4.3 μmol/l, p = 0.015) compared to patients who did not use these supplements. During the study no significant drop-in of nonstudy folic acid in the control group was encountered as plasma folic acid and homocysteine levels in this group remained unaltered (data not shown).

Folate status at inclusion was unrelated to the age of the

patients. However, vitamin B₁₂ level decreased with age ($r = -0.87$, $p = 0.035$). After correction for age and gender, the strongest predictors of plasma homocysteine level were levels of serum folate ($r = 0.31$, $p < 0.001$), vitamin B₁₂ level ($r = 0.22$, $p < 0.001$) and creatinine clearance ($r = 0.21$, $p < 0.001$). In the folic acid-treated group, plasma homocysteine level at three months' follow-up decreased, with 18% going from 12.0 ± 4.8 to 9.4 ± 3.5 $\mu\text{mol/l}$ ($p < 0.001$), and serum folate levels increased from 17 ± 7 to 33 ± 6 nmol/l ($p < 0.001$). These values persisted at six months' follow-up and did not change significantly. The decrement of plasma homocysteine was dependent on baseline values. In the lowest quartile, the percentage decrement was 9.6%, and in the third and fourth quartile this percentage increased up to 21% and 28%, respectively.

A total of 21 patients with asymptotically low vitamin B₁₂ levels (<120 pmol/l) were encountered: 11 patients in the folic acid-treated group and 10 control patients. When repeated measurements showed persistence of low vitamin B₁₂ levels, the general physician was given the advice to substitute cyanocobalamine i.m. Substitution occurred in 7 patients in both groups (14 total).

Homocysteine level at baseline was correlated with the log linear of serum folate with $r = -0.405$ ($p < 0.001$) and of vitamin B₁₂ with $r = -0.249$ ($p < 0.001$). After six months of treatment the relation of homocysteine level with serum folate was not significant anymore, but the correlation with vitamin B₁₂ persisted ($r = -0.204$, $p = 0.004$).

Fibrinogen and HS-CRP. During follow-up, fibrinogen as well as HS-CRP levels in both groups did not change significantly. The HS-CRP measurements were only performed in 158 cases included after April 2000 and were therefore excluded from the overall analysis. Yet, in cases where the data were available, HS-CRP levels were related to fibrinogen levels.

Clinical events and other follow-up data. After inclusion, 24 patients withdrew from the study (12 in each group), but were followed according to the protocol and were included in the analysis on an intention-to-treat basis. No patients were lost to follow-up.

Clinical cardiovascular events were evenly distributed in both treatment arms. In total, 37 (12.3%) clinical events in the folic acid group and 33 (11.2%) in the control group were observed, in 31 (10.3%) and 28 (9.6%) patients, respectively (Table 2). The time to the first clinical event is depicted in the Kaplan-Meier curve in Figure 1. No statistically significant difference existed between both groups (log-rank test, $p = 0.85$). This was also found for the patients in the highest quartile of plasma homocysteine (>13.7 $\mu\text{mol/l}$) (log-rank test, $p = 0.86$) (Fig. 1). Hospitalization for other reasons (which was not a part of the composite end point) was seen 95 (31.7%) times in the folic-acid group and 66 (22.5%) times in the control group and were mainly due to cataract or orthopedic surgery in this relative elderly population. Unstable angina or what we nowadays consider as non-ST-elevation acute coronary

Table 2. Cardiovascular Mortality and Morbidity

	Folic Acid	Control
Death (due to any reason)	12	14
Cardiovascular death	7	9
Noncardiovascular death	5	5
MI	3	4
PCI	6	3
CABG	4	3
CVA/TIA	4	3
Other vascular surgery	9	6
Death and/or any cardiovascular event	37	33

As per clinical event category, the number of patients with at least one event is presented. No double counting of patients for multiple events within a category was performed. Between categories, patients are double-counted in case of multiple events (e.g., a patient suffering two recurrent myocardial infarctions followed by a CABG is counted once in both categories).

Abbreviations as in Table 1.

syndrome (which was also not a part of the primary end point) was evenly distributed between both groups (11 cases in each group).

Survival analysis. Because no effect of folic acid intervention was observed, further analyses were done considering the study group as a whole. Four clinical parameters were dominant: age, history of diabetes, concomitant peripheral-vascular, and cerebrovascular disease. Three laboratory parameters were significant risk factors: calculated creatinine clearance, fibrinogen, and homocysteine. None of the on-treatment lipid values including apoA and apoB were found to be predictive.

Top-risk quartile analyses for the respective parameters, corrected for diabetes, gender, prior MI, PCI, and CABG, are depicted in Table 3. The respective Kaplan-Meier curves are shown in Figure 2. Nevertheless, in a multivariate model, corrected for the above-mentioned five clinical parameters, homocysteine and fibrinogen levels disappeared as significant risk factors when creatinine clearance level was introduced.

DISCUSSION

In this study, treatment with folic acid did not result in reduction of recurrence of cardiovascular events. The study was performed with the hypothesis that folic acid could have an additive effect in secondary prevention. This hypothesis was tested in a relatively high-risk population of patients with stable CAD with already optimally regulated lipid levels in addition to other well-known secondary prevention measures. At entry, most patients had already been on statin therapy for more than three years. The risk reduction thus pursued may have overshadowed the potential beneficial effects of folic acid on progression of atherosclerosis.

This study was designed on the basis of previous epidemiological evidence. In the past, much enthusiasm has been provoked by mostly cross-sectional and case-control studies (1-6,13-16), though with time, most prospective studies indicated less or absent association between homocysteine levels and cardiovascular risk. Therefore, it is reasonable to examine different meta-analyses of prospective studies that

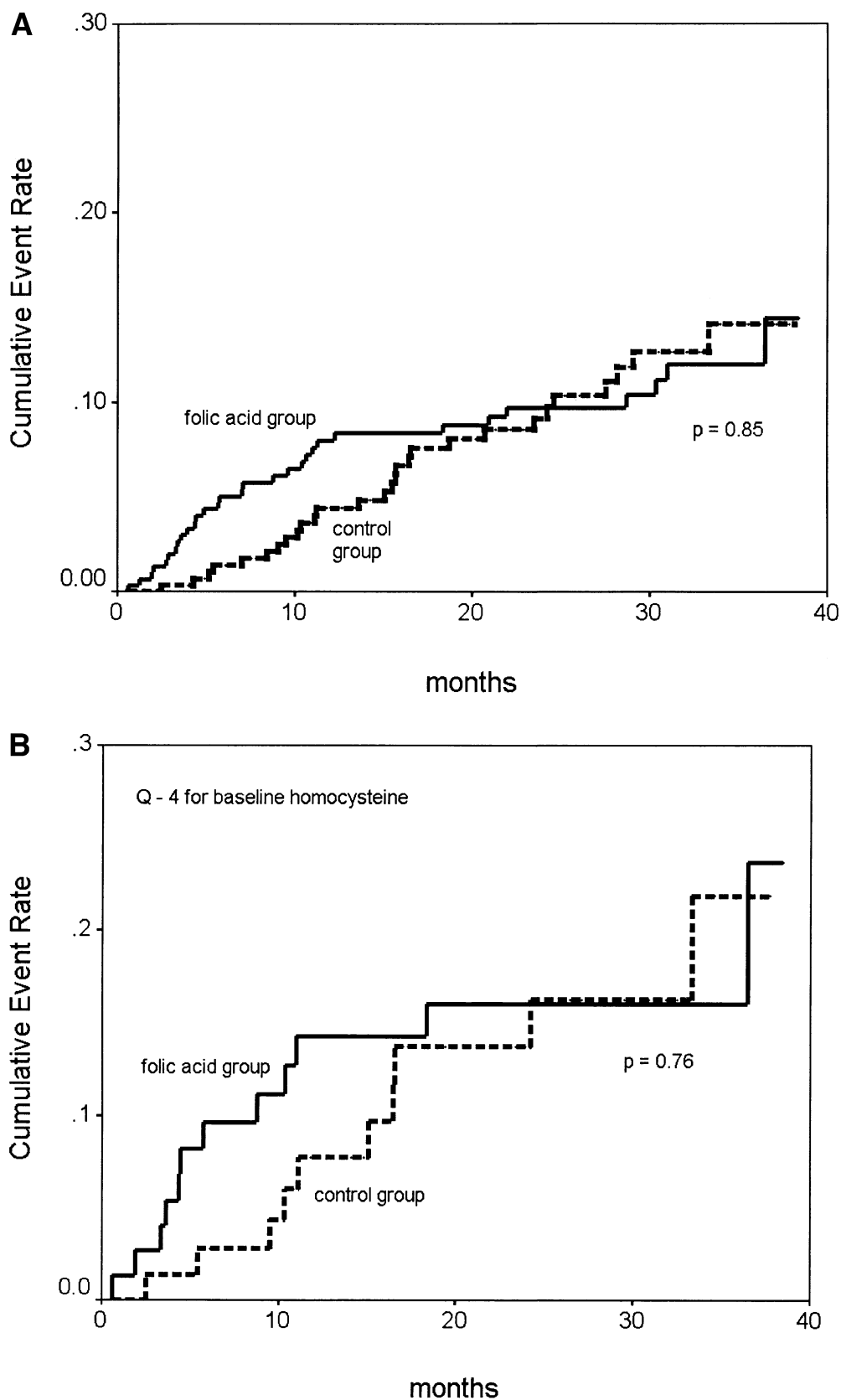


Figure 1. Cumulative event rate according to randomization to folic acid treatment (solid line) and control (dashed line). (A) Analysis for the entire group and (B) for patients with the highest quartile (Q-4) for baseline plasma homocysteine level ($\geq 13.7 \mu\text{mol/l}$). Probability value by log-rank statistic.

Table 3. Contribution of Fibrinogen, Creatinine Clearance, and Homocysteine to the Survival Model

Factor	RR*	95% CI	P Value
Creatinine clearance	2.13	1.23–3.69	0.007
Fibrinogen	2.10	1.23–3.56	0.007
Homocysteine	1.96	1.15–3.33	0.014

The survival model includes adjustment for diabetes, gender, prior myocardial infarction, percutaneous coronary intervention, and coronary artery bypass graft. *The relative risk of experiencing an event per unit time for patients in the top-risk quartile (Q_4 for fibrinogen and homocysteine; Q_4 for creatinine clearance) to patients in the lower three risk quartiles (Q_{1-3} for fibrinogen and homocysteine, Q_{2-4} for creatinine clearance) of the specified factor. The top-risk quartiles were ≥ 5 g/l for fibrinogen, ≥ 13.7 $\mu\text{mol/l}$ for homocysteine, and <60 ml/min for creatinine clearance.

CI = confidence interval; RR = relative risk.

have recently been published. One of those meta-analyses suggests that the odds ratio (OR) for ischemic heart disease for a 5 $\mu\text{mol/l}$ increase in serum homocysteine was 1.32 (95% confidence interval [CI] 1.19 to 1.45) (15). Another meta-analysis suggests that a decrease in homocysteine level of about 3 $\mu\text{mol/l}$ was associated with an 11% (OR 0.89; 95% CI 0.83 to 0.96) lower risk of ischemic heart disease and 19% (OR 0.81, 95% CI 0.69 to 0.95) lower stroke risk in healthy populations (16). It is important to note that most of the studies considered in the meta-analyses primarily included those performed in previously healthy populations so as to avoid confounding by disease. Only a few studies concern a prospective observation of mortality in patients with CAD (1–4).

In the observational study of Nygard et al. (2) a wide dose-response relation was observed within the range of homocysteine values, from about 5 $\mu\text{mol/l}$ to more than 20 $\mu\text{mol/l}$. The investigators calculated an adjusted mortality ratio of 1.9 for patients with total homocysteine levels between 9.0 and 14.9 $\mu\text{mol/l}$ as compared to those with values below 9 $\mu\text{mol/l}$. In our study, at six months, mean homocysteine levels were 12.2 ± 3.8 $\mu\text{mol/l}$ in the control group and 9.4 ± 3.7 $\mu\text{mol/l}$ in the folic acid-treated group, respectively. Consequently, on the basis of these figures one can expect to find a RR of about 0.5 (1/1.9) with folic acid intervention in patients with stable CAD.

Moreover, in another case-control study (3) the RR among men with known heart disease at baseline was, after adjustment for known risk factors, 2.23 (95% CI 1.03 to 4.85) in the highest serum homocysteine quintile compared with the lowest; yet the corresponding RR was only 0.90 (95% CI 0.51 to 1.60) among the men free of heart disease at baseline. This finding could indicate that the prognostic role of homocysteine is more pronounced in populations with existing heart disease compared to initially healthy populations.

Indeed, in our study it could be confirmed that plasma homocysteine levels at entry correlated with the risk of recurrent events, yet we did not observe decrement of risk with reduction of plasma homocysteine with folic acid, even when we only considered patients with the highest quartile of plasma homocysteine.

What are the existing data concerning folic acid intervention in coronary heart disease in terms of hard clinical end points? One recent study concerns a population undergoing PCI (11). A combination treatment of folic acid, vitamin B₁₂, and pyridoxine decreases the rate of restenosis and the need for revascularization. Nonetheless, no difference was observed in the rate of death from cardiac causes or nonfatal MI. Because of intervention of a vitamin combination, the decrement of homocysteine in this study was more pronounced, from 11.1 ± 4.3 to 7.2 ± 2.4 $\mu\text{mol/l}$.

Another placebo-controlled intervention with folic acid 5 mg/day in a population with ischemic heart disease was recently presented (17). The study was performed in a comparable population as our study, with 1,882 patients observed for a median of 1.7 years. Treatment reduced homocysteine level from 11.2 ± 6.9 $\mu\text{mol/l}$ to 9.7 ± 5.3 $\mu\text{mol/l}$, which is largely comparable to our observation. However, this study also did not observe reduction in risk in composite end point consisting of nonfatal MI, cardiovascular death, or unplanned revascularization (RR 0.97, 95% CI 0.72 to 1.29). Thus, these results are highly comparable to our own observations.

One potential limitation of our study concerns the inclusion of patients already using vitamin B-containing supplements on their own initiative. This is done on purpose for extrinsic validity reasons as it is well appreciated that a substantial proportion of the population uses the medication on a (semi)regular basis; in our study the percentage of users was about 15%. Yet when we dismissed these patients in the overall analysis, the results remained more or less unaltered. One limitation other ongoing studies might encounter elsewhere is the folic acid food fortification program. However, in the Netherlands this program was not introduced until fairly recently.

Other observations in this study are worth mentioning. In accord with the published data, in our study it was observed that homocysteine levels are related to age, folate status, and vitamin B₁₂ levels (18). In addition, it could be confirmed that the amount of reduction of plasma homocysteine is dependent on homocysteine level at baseline; the higher the baseline of homocysteine, the more reduction can be achieved. This reduction can be up to 28% in the highest quartile of plasma homocysteine. Yet this phenomenon could also be explained by the regression to the mean. Furthermore, we observed that, while patients are on folic acid treatment, homocysteine is no longer related to the height of serum folate, but that relations of homocysteine with vitamin B₁₂ levels persist. Thus, if higher homocysteine reduction is pursued—for instance, with food fortification—it is probably favorable not only to supplement folic acid but also vitamin B₁₂. This observation is a confirmation of a recent suggestion (19).

The matter of causality of homocysteine in atherosclerosis remains a highly debatable issue. Although we and others observed a relation between homocysteine level at entry of the study and prognosis, other laboratory parameters seem

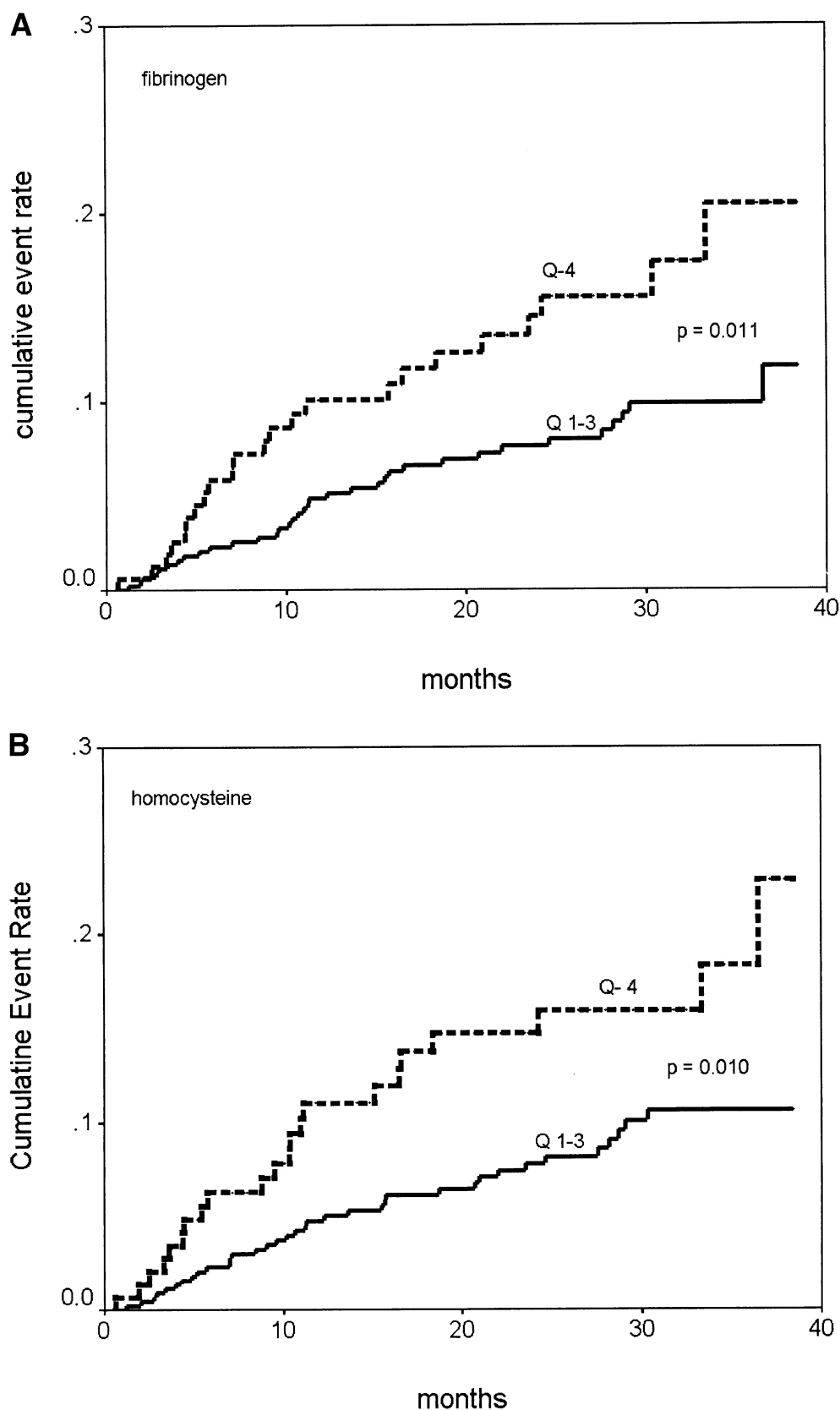


Figure 2. Cumulative rate of first cardiovascular event for two markers. Kaplan-Meier cumulative event rates for top-risk quartile (dashed line) versus lower three risk quartiles (solid line) are shown for fibrinogen (A) and homocysteine (B). The Q-4 is top-risk quartile for fibrinogen (≥ 5 g/l) and homocysteine (≥ 13.7 $\mu\text{mol/l}$). Value by log-rank statistic.

to be more prominent. One of these parameters is creatinine clearance. In most observational studies concerning homocysteine, only creatinine level was measured. We are more comfortable with calculated creatinine clearance as a measure of renal function. As it turns out, we observed a strong relation of creatinine clearance with recurrent events as renal function probably reflects the extent of cardiovascular disease. In addition, renal function plays a key role in the metabolism of homocysteine. Thus, it is conceivable that, among others, homocysteine is also a marker of renal function and that modulating this marker does not influence cardiovascular prognosis.

Another marker with a well-known prognostic characteristic is fibrinogen (20,21). In addition, fibrinogen is an independent predictor for subsequent acute coronary syndromes in patients with stable angina (22). Moreover, fibrinogen not only seems to be linked to thrombosis, but also to inflammation (23). Nowadays, both phenomena are considered highly relevant in the mechanism of disease of atherosclerosis (24). In our study we could confirm the relation between fibrinogen levels and subsequent events. Overall, in a multifactorial survival model with adjustments for clinical factors, the most predictive laboratory parameters in our study were, in order of significance, creatinine clearance, plasma fibrinogen, and homocysteine.

Study limitations. The current open-label study was powered for a 50% reduction of the event-rate under folic acid on the basis of observational studies in patients with established CAD (1-4). However, in a recent meta-analysis (yet predominantly including studies of patients without preexisting vascular disease) it was calculated that lowering homocysteine by 3 mmol/l from current levels reduces the risk of ischemic heart disease by 16% (11% to 20%) (15). Of course, our study is underpowered to see such a relatively small effect. In our investigation (with 2.6 μ mol/l lowering of mean plasma homocysteine) the event rates at 24 months were 9.2% and 9.7% in the placebo and folic acid groups, respectively. The 95% CI of RR of folic acid treatment ranged from 0.63 to 1.75, meaning that we cannot exclude a positive effect of folic acid treatment as great as 37% on risk reduction. Additionally, the time frame of two years necessary to see any effect may be of importance. Nevertheless, it is fair to say that one cannot expect major risk-reduction with folic acid substitution in the setting of secondary prevention within the time frame of two years. This is in agreement with recent findings (17). Nevertheless, we should not overinterpret the results of our study, as a lack of evidence is not the same as an evidence of lack of effect. We therefore have to await the results of ongoing trials in larger populations and with a longer follow-up time, such as the NORVIT, VITATOPS, and SEARCH trials, before one can support the routine use of folic acid supplementation in patients with ischemic heart disease.

Conclusions. In conclusion, low-dose folic acid in addition to statin therapy does not seem to affect the progress of cardiovascular disease within two years in terms of hard

clinical end points in patients with stable CAD. Homocysteine level is tightly linked to creatinine clearance. In addition, creatinine clearance can be interpreted as a mirror of the extent of atherosclerotic disease. Thus, homocysteine levels might, among others, be a marker of atherosclerotic disease. This study therefore does not seem to support the routine use of folic acid in patients with stable CAD.

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